Systems of two linear equations in two variables — Part 2 answer key

#	ID	Answer	Difficulty
2.1	016627-DC	A	Medium
2.2	572fa74d-4b79-415a-8241-a2848bbba074	A	Medium
2.3	09372-DC	C	Medium
2.4	181b5ada-d520-4724-9962-4d99a5d79e8f	D	Medium
2.5	cb6510bf-691c-4361-8c92-372e7946de91	B	Medium
2.6	e8d1c0cc-b06d-4425-92fb-3beba3af4800	80	Medium
2.7	7f5e8f0d-75c3-4c2d-ac2a-90f3c586f44c	C	Medium
2.8	a9ea688b-0693-4bef-8150-b731185309a2	C	Medium
2.9	025580-DC	?	Medium
2.10	034cf40e-98a7-4817-a3d0-a76cae0faf07	D	Medium
2.11	f02c7424-034b-4774-b41c-847249027a2e	C	Medium
2.12	04476-DC	?	Medium
2.13	10551-DC	D	Medium
2.14	87828f3a-da5a-4eeb-9cec-04f0ff9e1f97	C	Medium
2.15	a87ec507-656c-4bf5-ad90-09107b7dcf8b	B	Medium
2.16	33e5c79f-6eb4-44bc-917c-7090df8d4158	A	Medium
2.17	42dab414-d361-4e77-a550-5863631c999e	B	Medium
2.18	aa6865b9-fa65-43d3-bbf6-6e17250dc546	18	Medium
2.19	765f83e6-6d57-4694-82d5-9b2180d80565	D	Medium
2.20	11158-DC	C	Medium

2.21	b992e5d3-e84d-4c00-86c0-75f34b0c6f14	C	Medium
2.22	ab4fb25b-f569-45af-8536-c3264af6ff16	D	Hard
2.23	e124013e-471c-4f28-a3b4-212a4fd25c4e	D	Hard
2.24	bc24a37b-b601-44de-a222-60974e9db0fb	B	Hard
2.25	15e8ec7c-ca9d-47c1-ab6b-a40b86f74651	1677	Hard
2.26	c240c048-e272-462b-b645-1751de050cdd	D	Hard
2.27	031d354d-96ee-46e5-b453-c0be0ad459d5	B	Hard
2.28	e07c6477-a704-4fb6-8142-4f45390607dc	87	Hard
2.29	673c8b0c-840e-41ac-bc4e-82b2f8b2a747	B	Hard
2.30	060e402c-37f7-47bb-8481-4930b50fde96	C	Hard
2.31	bf3d7c38-346b-4b31-8065-9d69383f8a13	A	Hard
2.32	e9d39168-6488-45c7-828e-71039f10ceca	300	Hard
2.33	f6ec6740-9074-4a1f-a62d-cac86fd126d8	A	Hard
2.34	3bd8d2aa-c0a4-4f79-8d4e-5eb3b5c814f0	10	Hard
2.35	64207213-df09-436f-aeeb-a588aafa2b18	-34	Hard
2.36	029346-DC	B	Hard
2.37	029548-DC	D	Hard
2.38	82f7e038-e8ce-4d33-b043-0d0d68435a80	20	Hard
2.39	070616-DC	?	Hard
2.40	034116-DC	B	Hard
2.41	7f109a34-c477-4f2b-9704-3ccd5306ad84	D	Hard
2.42	5c0525ca-3f74-406b-82bd-232050622e14	D	Hard
2.43	96dd952c-8f39-4328-a087-4989624fb279	35	Hard
2.44	b07eee5e-2b48-42ca-9bdf-9cdd01089b62	B	Hard
2.45	9961525f-7f16-4e8c-9aef-754d7403a0cf	D	Hard
2.46	8b3e73a1-172a-4ae8-af4d-a8d63b197417	C	Hard

2.47	885fbade-61fe-4b3d-b976-1a5abfa9ae0c	D	Hard
2.48	99370f6d-8788-4824-a8aa-20c268f5a86c	B	Hard
2.49	f55f83d0-fd97-488f-9f5c-014bc1e34d9c	.2857, 2/7	Hard
2.50	35175215-d90c-49a0-927c-7f816dae6a5d	D	Hard
2.51	872b58e5-a9ea-488a-917d-0bdaaf2a4d73	3.5, 7/2	Hard
2.52	37c0d421-25f7-4ad6-99d7-8c81927bd65c	25	Hard
2.53	f1fcd45c-00d6-4477-bbcd-75ebc3cfc227	-28	Hard
2.54	070615-DC	?	Hard
2.55	05268-DC	?	Hard
2.56	022887-DC	D	Hard
2.57	05435-DC	D	Hard
2.58	d384e4e9-870d-4026-ba15-b7eb23e0c5fc	3	Hard
2.59	386cf16e-ddc8-4613-85b9-de02ba844834	8	Hard
2.60	425d30d9-246b-4058-b45c-2e4168f92a0f	1.8, 9/5	Hard
2.61	c4e7a98c-fecd-45d1-ad5b-69c978af6bf7	A	Hard
2.62	3265ed42-e3f1-44ac-9cb6-a8b55c776c51	11	Hard
2.63	5c09d547-9da9-4b9a-a436-ddc8c0c70764	6	Hard

Rationales

▼ 2.1 — A

Choice A is correct. The solution to a system of equations is the coordinates of the intersection point of the graphs of the equations in the xy-plane. Based on the graph, the solution to the given system of equations is best approximated as $(\frac{3}{2},0)$. In the xy-plane, the graph of $y=0$ is a horizontal line on which every y-coordinate is 0, and the graph of $x=\frac{3}{2}$ is a vertical line on which every x-coordinate is $\frac{3}{2}$. These graphs intersect at the

point $(\frac{3}{2},0)$. Therefore, the system of equations in choice A has the same solution as the given system.

Choices B, C, and D are incorrect. If graphed in the xy-plane, these choices would intersect at the points $(0,\frac{3}{2})$, $(1,0)$, and $(0,1)$, respectively, not $(\frac{3}{2},0)$.

▼ 2.2 — A

Choice A is correct. The given system of linear equations can be solved by the substitution method. The first equation in the given system of equations defines y as $6x + 16$. Substituting $6x + 16$ for y in the second equation of the given system of equations yields $-7x - (6x + 16) = 36$. Applying the distributive property on the left-hand side of this equation yields $-7x - 6x - 16 = 36$, or $-13x - 16 = 36$. Adding 16 to both sides of this equation yields $-13x = 52$. Dividing both sides of this equation by -13 yields $x = -4$. Substituting -4 for x in the first equation of the given system of equations, $y = 6x + 16$, yields $y = 6(-4) + 16$, or $y = -8$. Therefore, the solution (x,y) to the given system of equations is $(-4, -8)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.3 — C

Choice C is correct. Let x represent the number of 2-person tents and let y represent the number of 4-person tents. It is given that the total number of tents was 60 and the total number of people in the group was 202. This situation can be expressed as a system of two equations, $x+y=60$ and $2x+4y=202$. The first equation can be rewritten as $y=-x+60$. Substituting $-x+60$ for y in the equation $2x+4y=202$ yields $2x+4(-x+60)=202$. Distributing and combining like terms gives $-2x+240=202$. Subtracting 240 from both sides of $-2x+240=202$ and then dividing both sides by -2 gives $x=19$. Therefore, the number of 2-person tents is 19.

Alternate approach: If each of the 60 tents held 4 people, the total number of people that could be accommodated in tents would be 240. However, the actual number of people who

slept in tents was 202. The difference of 38 accounts for the 2-person tents. Since each of these tents holds 2 people fewer than a 4-person tent, $\frac{38}{2} = 19$ gives the number of 2-person tents.

Choice A is incorrect. This choice may result from assuming exactly half of the tents hold 2 people. If that were true, then the total number of people who slept in tents would be $2(30) + 4(30) = 180$; however, the total number of people who slept in tents was 202, not 180. Choice B is incorrect. If 20 tents were 2-person tents, then the remaining 40 tents would be 4-person tents. Since all the tents were filled to capacity, the total number of people who slept in tents would be $2(20) + 4(40) = 40 + 160 = 200$; however, the total number of people who slept in tents was 202, not 200. Choice D is incorrect. If 18 tents were 2-person tents, then the remaining 42 tents would be 4-person tents. Since all the tents were filled to capacity, the total number of people who slept in tents would be $2(18) + 4(42) = 36 + 168 = 204$; however, the total number of people who slept in tents was 202, not 204.

▼ 2.4 — D

Choice D is correct. Adding the second equation to the first equation in the given system of equations yields $3y - 3y = 4x + 9x + 17 - 23$, or $0 = 13x - 6$. Adding 6 to each side of this equation yields $6 = 13x$. Multiplying each side of this equation by 3 yields $18 = 39x$. Therefore, the value of $39x$ is 18.

Choice A is incorrect. This is the value of $-39x$, not $39x$.

Choice B is incorrect. This is the value of $-13x$, not $39x$.

Choice C is incorrect. This is the value of $13x$, not $39x$.

▼ 2.5 — B

Choice B is correct. It's given by the first equation in the system that $y = -\frac{1}{5}x$. Substituting $-\frac{1}{5}x$ for y in the second equation in the system, $y = \frac{1}{7}x$, yields $-\frac{1}{5}x = \frac{1}{7}x$. Adding $-\frac{1}{5}x$ to both sides of this equation yields $0 = \frac{1}{7}x + \frac{1}{5}x$, which is equivalent to $0 = \frac{5}{35}x + \frac{7}{35}x$, or $0 = \frac{12}{35}x$. Multiplying both sides of this equation by $\frac{35}{12}$ yields $0 = x$. Therefore, the value of x is 0.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.6 — 80

The correct answer is 80. Subtracting the second equation in the given system from the first equation yields $24x + y - 6x + y = 48 - 72$, which is equivalent to $24x - 6x + y - y = -24$, or $18x = -24$. Dividing each side of this equation by 3 yields $6x = -8$. Substituting -8 for $6x$ in the second equation yields $-8 + y = 72$. Adding 8 to both sides of this equation yields $y = 80$.

Alternate approach: Multiplying each side of the second equation in the given system by 4 yields $24x + 4y = 288$. Subtracting the first equation in the given system from this equation yields $24x + 4y - 24x + y = 288 - 48$, which is equivalent to $24x - 24x + 4y - y = 240$, or $3y = 240$. Dividing each side of this equation by 3 yields $y = 80$.

▼ 2.7 — C

Choice C is correct. Adding $9x$ to both sides of the first equation in the given system yields $y = 9x + 13$. Substituting the expression $9x + 13$ for y in the second equation in the given system yields $5x = 29x + 13$. Distributing the 2 on the right-hand side of this equation yields $5x = 18x + 26$. Subtracting $18x$ from both sides of this equation yields $-13x = 26$. Dividing both sides of this equation by -13 yields $x = -2$. Substituting -2 for x in the equation $y = 9x + 13$ yields $y = 9\text{-}2 + 13$, or $y = -5$. Therefore, the solution x,y to the given system of equations is $-2, -5$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the solution y,x, not x,y, to the given system of equations.

▼ 2.8 — C

Choice C is correct. A system of two linear equations in two variables, x and y, has no solution if the graphs of the lines represented by the equations in the xy-plane are distinct and parallel. The graphs of two lines in the xy-plane represented by equations in slope-intercept form, $y = mx + b$, where m and b are constants, are parallel if their slopes, m, are the same and are distinct if their y-coordinates of the y-intercepts, b, are different. In the equations $y = 16x + 3$ and $y = 16x + 19$, the values of m are each 16, and the values of b are 3 and 19, respectively. Since the slopes of these lines are the same, and the y-coordinates

of the *y*-intercepts are different, it follows that the system of linear equations in choice C has no solution.

Choice A is incorrect. The lines represented by the equations in this system are a vertical line and a horizontal line. Therefore, this system has a solution, $3, 5$, rather than no solution.

Choice B is incorrect. The two lines represented by these equations have different slopes and the same *y*-coordinate of the *y*-intercept. Therefore, this system has a solution, $0, 6$, rather than no solution.

Choice D is incorrect. The two lines represented by these equations are a horizontal line and a line with a slope of 5 that have the same *y*-coordinate of the *y*-intercept. Therefore, this system has a solution, $0, 5$, rather than no solution.

▼ 2.9 — ?

The correct answer is 7. Subtracting the second equation from the first equation eliminates the variable y.

$$\begin{aligned} 5x + 3y &= 38 \\ -(x + 3y &= 10) \\ 4x &= 28 \end{aligned}$$

Dividing both sides of the resulting equation by 4 yields x = 7.

▼ 2.10 — D

Choice D is correct. The first equation in the given system of equations defines *y* as $3x + 9$. Substituting $3x + 9$ for *y* in the second equation in the given system of equations yields $3(3x + 9) = 8x - 6$. Applying the distributive property on the left-hand side of this equation yields $9x + 27 = 8x - 6$. Subtracting $8x$ from both sides of this equation yields $x + 27 = -6$. Subtracting 27 from both sides of this equation yields $x = -33$. Substituting -33 for *x* in the first equation of the given system of equations yields $y = 3(-33) + 9$, or $y = -90$. Substituting -33 for *x* and -90 for *y* into the expression $x - y$ yields $-33 - (-90)$, or 57.

Choice A is incorrect. This is the value of $x + y$, not $x - y$.

Choice B is incorrect. This is the value of *x*, not $x - y$.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.11 — C

Choice C is correct. Let x represent the number of children in a whale-watching tour group. Let y represent the number of adults in this group. Because it's given that 21 people are in a group and the group consists of adults and children, it must be true that $x + y = 21$. Since the company's revenue is 60 dollars per child, the total revenue from x children in this group was $60x$ dollars. Since the company's revenue is 80 dollars per adult, the total revenue from y adults in this group was $80y$ dollars. Because it's given that the total revenue for this group was $1{,}440$ dollars, it must be true that $60x + 80y = 1{,}440$. The equations $x + y = 21$ and $60x + 80y = 1{,}440$ form a linear system of equations that can be solved to find the value of x, which represents the number of children in the group, using the elimination method. Multiplying both sides of the equation $x + y = 21$ by 80 yields $80x + 80y = 1{,}680$. Subtracting $60x + 80y = 1{,}440$ from $80x + 80y = 1{,}680$ yields $(80x + 80y) - (60x + 80y) = 1{,}680 - 1{,}440$, which is equivalent to $80x - 60x + 80y - 80y = 240$, or $20x = 240$. Dividing both sides of this equation by 20 yields $x = 12$. Therefore, 12 people in the group were children.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of adults in the group, not the number of children in the group.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.12 — ?

The correct answer is 30. Let x represent the number of correct answers from the player and y represent the number of incorrect answers from the player. Since the player answered 40 questions in total, the equation $x + y = 40$ represents this situation. Also, since the score is found by subtracting the number of incorrect answers from twice the number of correct answers and the player received a score of 50, the equation $2x - y = 50$ represents this situation. Adding the equations in the system of two equations together yields $(x + y) + (2x - y) = 40 + 50$. This can be rewritten as $3x = 90$. Finally, solving for x by dividing both sides of the equation by 3 yields $x = 30$.

▼ 2.13 — D

Choice D is correct. Adding the corresponding sides of the two equations eliminates y and yields $x = 6$, as shown.

$$\begin{array}{r} \frac{1}{2}y = 4 \\ x - \frac{1}{2}y = 2 \\ \hline x + 0 = 6 \end{array}$$

If (x, y) is a solution to the system, then (x, y) satisfies both equations in the system and any equation derived from them. Therefore, $x = 6$.

Choices A, B, and C are incorrect and may be the result of errors when solving the system.

▼ 2.14 — C

Choice C is correct. It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions when the two linear equations are equivalent. Dividing both sides of the given equation by 5 yields $-3x + 5y = 13$. Dividing both sides of choice C by 4 also yields $-3x + 5y = 13$, so choice C is equivalent to the given equation. Thus, choice C could be the second equation in the system.

Choice A is incorrect. The system consisting of this equation and the given equation has one solution, not infinitely many solutions.

Choice B is incorrect. The system consisting of this equation and the given equation has one solution, not infinitely many solutions.

Choice D is incorrect. The system consisting of this equation and the given equation has no solution, not infinitely many solutions.

▼ 2.15 — B

Choice B is correct. The second equation in the given system is $y = 9x + 1$. Substituting $9x + 1$ for y in the first equation in the given system yields $8x + 9x + 1 = 5$, which is equivalent to $17x + 1 = 5$. Subtracting 1 from both sides of this equation yields $17x = 4$. Dividing both sides of this equation by 17 yields $x = \frac{4}{17}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.16 — A

Choice A is correct. A system of two linear equations in two variables, x and y, has no solution if the graphs of the lines represented by the equations in the xy-plane are distinct and parallel. The graphs of two lines in the xy-plane represented by equations in slope-intercept form, $y = mx + b$, where m and b are constants, are parallel if their slopes, m, are the same and are distinct if their y-coordinates of the y-intercepts, b, are different. In the equations $y = 6x + 3$ and $y = 6x + 9$, the values of m are each 6, and the values of b are 3 and 9, respectively. Since the slopes of these lines are the same and the y-coordinates of the y-intercepts are different, it follows that the system of linear equations in choice A has no solution.

Choice B is incorrect. The two lines represented by these equations are a horizontal line and a line with a slope of 10 that have the same y-coordinate of the y-intercept. Therefore, this system has a solution, $0, 10$, rather than no solution.

Choice C is incorrect. The two lines represented by these equations have different slopes and the same y-coordinate of the y-intercept. Therefore, this system has a solution, $0, 14$, rather than no solution.

Choice D is incorrect. The two lines represented by these equations are a vertical line and a horizontal line. Therefore, this system has a solution, $3, 10$, rather than no solution.

▼ 2.17 — B

Choice B is correct. It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions when the two linear equations are equivalent. Adding 6 to both sides of $y - 6 = \frac{4}{5}x + 5$ yields $y = \frac{4}{5}x + 5 + 6$. Applying the distributive property to this equation yields $y = \frac{4}{5}x + \frac{20}{5} + 6$, or $y = \frac{4}{5}x + 10$. Adding v to both sides of $y - v = \frac{4}{5}x + 10$ yields $y = \frac{4}{5}x + 10 + v$. Applying the distributive property to this equation yields $y = \frac{4}{5}x + \frac{40}{5} + v$, or $y = \frac{4}{5}x + 8 + v$. For the equations $y = \frac{4}{5}x + 10$ and $y = \frac{4}{5}x + 8 + v$ to be equivalent, it follows that $10 = 8 + v$. Subtracting 8 from both sides of this equation yields $2 = v$. Therefore, the value of v is 2.

Choice A is incorrect. If $v = 0$, the system of equations would have no solution, not infinitely many solutions.

Choice C is incorrect. If $v = 4$, the system of equations would have no solution, not infinitely many solutions.

Choice D is incorrect. If $v = 12$, the system of equations would have no solution, not infinitely many solutions.

▼ 2.18 — 18

The correct answer is 18. It's given by the second equation in the system that $3y = 11$. Substituting 11 for $3y$ in the first equation in the system, $x + 3y = 29$, yields $x + 11 = 29$. Subtracting 11 from both sides of this equation yields $x = 18$.

▼ 2.19 — D

Choice D is correct. The first equation in the given system is $y = 4x$. Substituting $4x$ for y in the second equation yields $5x - 3(4x) = -14$, which is equivalent to $5x - 12x = -14$, or $-7x = -14$. Dividing both sides of this equation by -7 yields $x = 2$. Substituting 2 for x in the equation $y = 4x$ yields $y = 4(2)$, or $y = 8$. Substituting 2 for x and 8 for y in the expression $x + y$ yields $2 + 8$, or 10.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $y - x$, not the value of $x + y$.

Choice C is incorrect. This is the value of y, not the value of $x + y$.

▼ 2.20 — C

Choice C is correct. Adding the given equations yields 9x + 9y = 162. Dividing each side of the equation 9x + 9y = 162 by 9 gives x + y = 18.

Choice A is incorrect and may result from incorrectly adding the equations. Choice B is incorrect and may result from conceptual or computational errors. Choice D is incorrect. This value is equivalent to y – x.

▼ 2.21 — C

Choice C is correct. It's given by the first equation in the system that $y = -\frac{1}{9}x$. Substituting $-\frac{1}{9}x$ for y in the second equation in the system yields $-\frac{1}{9}x = \frac{1}{2}x$. Multiplying the left-hand side of this equation by $\frac{2}{2}$ and the right-hand side by $\frac{9}{9}$ yields $-\frac{2}{18}x = \frac{9}{18}x$. Adding $\frac{2}{18}x$ to both sides of this equation yields $0 = \frac{11}{18}x$. Multiplying both sides of this equation by $\frac{18}{11}$ yields $x = 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.22 — D

Choice D is correct. It's given that the graphs of the given system of equations intersect at the point $4, y$. Therefore, $4, y$ is the solution to the given system. Multiplying the first equation in the given system by -2 yields $-2ax - 2by = -144$. Adding this equation to the second equation in the system yields $-2a + 6x + -2b + 2by = -144 + 56$, or $-2a + 6x = -88$. Since $4, y$ is the solution to the system, the value of a can be found by substituting 4 for x in this equation, which yields $-2a + 64 = -88$. Dividing both sides of this equation by 4 yields $-2a + 6 = -22$. Subtracting 6 from both sides of this equation yields $-2a = -28$. Dividing both sides of this equation by -2 yields $a = 14$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.23 — D

Choice D is correct. A system of linear equations can be solved by the elimination method. Multiplying the equation $-2x + 3y = 9$ by 2 yields $-4x + 6y = 18$. Adding this equation to the equation $4x - 6y = 18$ yields $0 = 36$, which has no solution. It follows that the system of linear equations consisting of $-2x + 3y = 9$ and $4x - 6y = 18$ has no solution.

Choice A is incorrect. This system of linear equations has infinitely many solutions, rather than no solution.

Choice B is incorrect. This system of linear equations has one solution, rather than no solution.

Choice C is incorrect. This system of linear equations has infinitely many solutions, rather than no solution.

▼ 2.24 — B

Choice C is correct. It's given that store A sells raspberries for $\$5.50$ per pint and blackberries for $\$3.00$ per pint, and a certain purchase of raspberries and blackberries at

store A would cost $\$37.00$. It's also given that store B sells raspberries for $\$6.50$ per pint and blackberries for $\$8.00$ per pint, and this purchase of raspberries and blackberries at store B would cost $\$66.00$. Let r represent the number of pints of raspberries and b represent the number of pints of blackberries in this purchase. The equation $5.50r + 3.00b = 37.00$ represents this purchase of raspberries and blackberries from store A and the equation $6.50r + 8.00b = 66.00$ represents this purchase of raspberries and blackberries from store B. Solving the system of equations by elimination gives the value of r and the value of b that make the system of equations true. Multiplying both sides of the equation for store A by 6.5 yields $5.50r6.5 + 3.00b6.5 = 37.006.5$, or $35.75r + 19.5b = 240.5$. Multiplying both sides of the equation for store B by 5.5 yields $6.50r5.5 + 8.00b5.5 = 66.005.5$, or $35.75r + 44b = 363$. Subtracting both sides of the equation for store A, $35.75r + 19.5b = 240.5$, from the corresponding sides of the equation for store B, $35.75r + 44b = 363$, yields $35.75r - 35.75r + 44b - 19.5b = 363 - 240.5$, or $24.5b = 122.5$. Dividing both sides of this equationby 24.5 yields $b = 5$. Thus, 5 pints of blackberries are in
this purchase.

Choices A and B are incorrect and may result from conceptual or calculation errors. Choice D is incorrect. This is the number of pints of raspberries, not blackberries, in the purchase.

▼ 2.25 — 1677

The correct answer is $1{,}677$. Adding the first equation to the second equation in the given system yields $x - 2 + x - 2 + -4y + 7 + 4y + 7 = 117 + 442$, or $2x - 2 = 559$. Multiplying both sides of this equation by 3 yields $6x - 2 = 1{,}677$. Therefore, the value of $6x - 2$ is $1{,}677$.

▼ 2.26 — D

Choice D is correct. A line in the xy-plane that passes through the points x_1, y_1 and x_2, y_2 has slope m, where $m = \frac{y_2 - y_1}{x_2 - x_1}$, and can be defined by an equation of the form $y - y_1 = mx - x_1$. One of the lines shown in the graph passes through the points $8, 0$ and $3, 4$. Substituting 8 for x_1, 0 for y_1, 3 for x_2, and 4 for y_2 in the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{4 - 0}{3 - 8}$, or $m = -\frac{4}{5}$. Substituting $-\frac{4}{5}$ for m, 8 for x_1 and 0 for y_1 in the equation $y - y_1 = mx - x_1$ yields $y - 0 = -\frac{4}{5}x - 8$, which is equivalent to $y = -\frac{4}{5}x + \frac{32}{5}$. Adding $\frac{4}{5}x$ to both sides of this equation yields $\frac{4}{5}x + y = \frac{32}{5}$. Multiplying both sides of this equation by -10 yields $-8x - 10y = -64$. Therefore, an equation of this line is $-8x - 10y = -64$. Similarly, the other line shown in the graph passes through the points $8, 0$ and $3, 2$. Substituting 8 for x_1, 0 for y_1, 3 for x_2, and 2

for y_2 in the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{2 - 0}{3 - 8}$, or $m = -\frac{2}{5}$. Substituting $-\frac{2}{5}$ for m, 8 for x_1, and 0 for y_1 in the equation $y - y_1 = mx - x_1$ yields $y - 0 = -\frac{2}{5}x - 8$, which is equivalent to $y = -\frac{2}{5}x + \frac{16}{5}$. Adding $\frac{2}{5}x$ to both sides of this equation yields $\frac{2}{5}x + y = \frac{16}{5}$. Multiplying both sides of this equation by 10 yields $4x + 10y = 32$. Therefore, an equation of this line is $4x + 10y = 32$. So, the system of linear equations represented by the lines shown is $4x + 10y = 32$ and $-8x - 10y = -64$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.27 — B

Choice B is correct. A system of two linear equations in two variables, x and y, has no solution when the lines in the xy-plane representing the equations are parallel and distinct. Two lines are parallel and distinct if their slopes are the same and their y-intercepts are different. The slope of the graph of the given equation, $3x = 36y - 45$, in the xy-plane can be found by rewriting the equation in the form $y = mx + b$, where m is the slope of the graph and $0, b$ is the y-intercept. Adding 45 to each side of the given equation yields $3x + 45 = 36y$. Dividing each side of this equation by 36 yields $\frac{1}{12}x + \frac{5}{4} = y$, or $y = \frac{1}{12}x + \frac{5}{4}$. It follows that the slope of the graph of the given equation is $\frac{1}{12}$ and the y-intercept is $0, \frac{5}{4}$. Therefore, the graph of the second equation in the system must also have a slope of $\frac{1}{12}$, but must not have a y-intercept of $0, \frac{5}{4}$. Multiplying each side of the equation given in choice B by $\frac{1}{4}$ yields $\frac{1}{12}x = y$, or $y = \frac{1}{12}x$. It follows that the graph representing the equation in choice B has a slope of $\frac{1}{12}$ and a y-intercept of $0, 0$. Since the slopes of the graphs of the two equations are equal and the y-intercepts of the graphs of the two equations are different, the equation in choice B could be the second equation in the system.

Choice A is incorrect. This equation can be rewritten as $y = \frac{1}{4}x$. It follows that the graph of this equation has a slope of $\frac{1}{4}$, so the system consisting of this equation and the given equation has exactly one solution, rather than no solution.

Choice C is incorrect. This equation can be rewritten as $y = \frac{1}{12}x + \frac{5}{4}$. It follows that the graph of this equation has a slope of $\frac{1}{12}$and a y-intercept of $0, \frac{5}{4}$, so the system consisting of this equation and the given equation has infinitely many solutions, rather than no solution.

Choice D is incorrect. This equation can be rewritten as $y = \frac{1}{36}x + \frac{5}{4}$. It follows that the graph of this equation has a slope of $\frac{1}{36}$, so the system consisting of this equation and the given

equation has exactly one solution, rather than no solution.

▼ 2.28 — 87

The correct answer is 87. It's given that in August, the car dealer completed 15 more than 3 times the number of sales the car dealer completed in September. Let x represent the number of sales the car dealer completed in September. It follows that $3x + 15$ represents the number of sales the car dealer completed in August. It's also given that in August and September, the car dealer completed 363 sales. It follows that $x + 3x + 15 = 363$, or $4x + 15 = 363$. Subtracting 15 from each side of this equation yields $4x = 348$. Dividing each side of this equation by 4 yields $x = 87$. Therefore, the car dealer completed 87 sales in September.

▼ 2.29 — B

Choice B is correct. A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the xy-plane are parallel and distinct. Lines represented by equations in standard form, $Ax + By = C$ and $Dx + Ey = F$, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation, meaning $\frac{D}{A} = \frac{E}{B}$; and the lines are distinct if the constants are not proportional, meaning $\frac{F}{C}$ is not equal to $\frac{D}{A}$ or $\frac{E}{B}$. The given equation, $y = 6x + 18$, can be written in standard form by subtracting $6x$ from both sides of the equation to yield $-6x + y = 18$. Therefore, the given equation can be written in the form $Ax + By = C$, where $A = -6$, $B = 1$, and $C = 18$. The equation in choice B, $-6x + y = 22$, is written in the form $Dx + Ey = F$, where $D = -6$, $E = 1$, and $F = 22$. Therefore, $\frac{D}{A} = \frac{-6}{-6}$, which can be rewritten as $\frac{D}{A} = 1$; $\frac{E}{B} = \frac{1}{1}$, which can be rewritten as $\frac{E}{B} = 1$; and $\frac{F}{C} = \frac{22}{18}$, which can be rewritten as $\frac{F}{C} = \frac{11}{9}$. Since $\frac{D}{A} = 1$, $\frac{E}{B} = 1$, and $\frac{F}{C}$ is not equal to 1, it follows that the given equation and the equation $-6x + y = 22$ are parallel and distinct. Therefore, a system of two linear equations consisting of the given equation and the equation $-6x + y = 22$ has no solution. Thus, the equation in choice B could be the second equation in the system.

Choice A is incorrect. The equation $-6x + y = 18$ and the given equation represent the same line in the xy-plane. Therefore, a system of these linear equations would have infinitely many solutions, rather than no solution.

Choice C is incorrect. The equation $-12x + y = 36$ and the given equation represent lines in the xy-plane that are distinct and not parallel. Therefore, a system of these linear equations would have exactly one solution, rather than no solution.

Choice D is incorrect. The equation $-12x + y = 18$ and the given equation represent lines in the xy-plane that are distinct and not parallel. Therefore, a system of these linear equations would have exactly one solution, rather than no solution.

▼ 2.30 — C

Choice C is correct. A system of two linear equations in two variables, x and y, has no solution if the graphs of the equations in the xy-plane are two distinct and parallel lines, which don't intersect. Two lines in the xy-plane are distinct and parallel if their slopes are the same and the y-coordinates of their y-intercepts are different. A linear equation can be written in slope-intercept form, $y = mx + b$, where m is the slope of the line in the xy-plane and b is the y-coordinate of the y-intercept. Choice C can be written in slope-intercept form by adding $7x$ to both sides of the equation, which gives $y = 7x + 21$. In the equations $y = 7x + 18$ and $y = 7x + 21$, the values of m are each 7, and the values of b are 18 and 21, respectively. Since the slopes of these lines are the same and the y-coordinates of the y-intercepts are different, it follows that choice C could be the second equation in the system.

Choice A is incorrect. Adding $21x$ to both sides of this equation gives $y = 21x + 54$. In the equations $y = 7x + 18$ and $y = 21x + 54$, the values of m aren't equal, so the slopes of these lines are different. Therefore, these two lines intersect and the corresponding system of equations has one solution.

Choice B is incorrect. Adding $21x$ to both sides of this equation gives $y = 21x + 18$. In the equations $y = 7x + 18$ and $y = 21x + 18$, the values of m aren't equal, so the slopes of these lines are different. Therefore, these two lines intersect and the corresponding system of equations has one solution.

Choice D is incorrect. Adding $7x$ to both sides of this equation gives $y = 7x + 18$. Since this equation is the same as the given equation, these two lines are not distinct. Therefore, the corresponding system of equations has infinitely many solutions.

▼ 2.31 — A

Choice A is correct. It's given that if line k is graphed in the xy-plane with line h, the result is the graph of a system of two linear equations that has no solution. It follows that line h and line k don't intersect. This means that line k and line h have the same slope and different y-intercepts. The graph shows that the y-intercept of line h is $0, -9$. The y-coordinate of the y-intercept of line k is the value of y when $x = 0$ in the equation defining line k. Substituting 0 for x in the equation $sx + 48y = t$ yields $s0 + 48y = t$, or $48y = t$. Dividing both sides of this

equation by 48 yields $y = \frac{t}{48}$. Therefore, the *y*-coordinate of the *y*-intercept of line *k* is $\frac{t}{48}$, so the *y*-intercept of line *k* is $0, \frac{t}{48}$. Since line *k* and line *h* have different *y*-intercepts, $\frac{t}{48}$ can't be equal to -9. It follows that *t* can't be equal to -948, or -432. Thus, -432 isn't a possible value of *t*.

Choice B is incorrect. This isn't a possible value of the *y*-coordinate of the *y*-intercept of line *k*, but it is a possible value of *t*.

Choice C is incorrect. This is the value of *s*, not a value of *t* that isn't possible.

Choice D is incorrect. If $t = 288$, then the *y*-intercept of line *k* is $0, \frac{288}{48}$, or $0, 6$, which is different from the *y*-intercept of line *h*, $0, -9$. Therefore, 288 is a possible value of *t*.

▼ 2.32 — 300

The correct answer is 300. It's given that the graph of line *g* is shown in the *xy*-plane and that line *k* is defined by $165x + py = w$, where *p* and *w* are constants. It's also given that if line *k* is graphed in this *xy*-plane, the resulting system of two linear equations will have infinitely many solutions. A system of two linear equations has infinitely many solutions when the two linear equations are equivalent. Therefore, line *g* and line *k* have the same slope and the same *y*-intercept. The slope of a line that passes through the points (x_1, y_1) and (x_2, y_2) can be calculated using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The graph shows that line *g* passes through the points $(-10, 12)$ and $(0, 1)$. Substituting $(-10, 12)$ and $(0, 1)$ for (x_1, y_1) and (x_2, y_2), respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{1 - 12}{0 - (-10)}$, or $m = \frac{-11}{10}$. An equation representing a linear relationship can be written in the form $y = mx + b$, where *m* is the slope and $(0, b)$ is the *y*-intercept of the line in the *xy*-plane. Since line *g* passes through the point $(0, 1)$, it follows that $b = 1$. Thus, the equation of line *g* is $y = -\frac{11}{10}x + 1$. For line *k*, the equation $165x + py = w$ can be rewritten in $y = mx + b$ form by isolating *y*. Subtracting $165x$ from both sides of this equation yields $py = -165x + w$. Dividing both sides of this equation by *p* yields $y = \frac{-165}{p}x + \frac{w}{p}$. It follows that in the *xy*-plane, line *k* has a slope of $\frac{-165}{p}$ and $0, \frac{w}{p}$ is the *y*-intercept. Since line *g* and line *k* have the same slope and the same *y*-intercept, it follows that $-\frac{165}{p} = -\frac{11}{10}$ and $\frac{w}{p} = 1$. Multiplying both sides of $-\frac{165}{p} = -\frac{11}{10}$ by -10*p* yields $1{,}650 = 11p$. Dividing both sides of this equation by 11 yields $150 = p$. Substituting 150 for *p* in the equation $\frac{w}{p} = 1$ yields $\frac{w}{150} = 1$. Multiplying both sides of this equation by 150 yields $w = 150$. Substituting 150 for *p* and 150 for *w* in the expression $p + w$ yields $150 + 150$, which is equivalent to 300. Therefore, the value of $p + w$ is 300.

▼ 2.33 — A

Choice A is correct. A solution to a system of equations must satisfy each equation in the system. It follows that if an ordered pair x, y is a solution to the system, the point x, y lies on the graph in the xy-plane of each equation in the system. The only point that lies on each graph of the system of two linear equations shown is their intersection point $8, 2$. It follows that if a new graph of three linear equations is created using the system of equations shown and the graph of $x + 4y = -16$, this system has either zero solutions or one solution, the point $8, 2$. Substituting 8 for x and 2 for y in the equation $x + 4y = -16$ yields $8 + 42 = -16$, or $16 = -16$. Since this equation is not true, the point $8, 2$ does not lie on the graph of $x + 4y = -16$. Therefore, $8, 2$ is not a solution to the system of three equations. It follows that there are zero solutions to this system.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.34 — 10

The correct answer is 10. Solving by substitution, the given system of equations, where p is a constant, can be written so that the left-hand side of each equation is equal to $7r$. Subtracting 6 from each side of the first equation in the given system, $6 + 7r = pw$, yields $7r = pw - 6$. Adding $5w$ to each side of the second equation in the given system, $7r - 5w = 5w + 11$, yields $7r = 10w + 11$. Since the left-hand side of each equation is equal to $7r$, setting the the right-hand side of the equations equal to each other yields $pw - 6 = 10w + 11$. A linear equation in one variable, w, has no solution if and only if the equation is false; that is, when there's no value of w that produces a true statement. For the equation $pw - 6 = 10w + 11$, there's no value of w that produces a true statement when $pw = 10w$. Therefore, for the equation $pw - 6 = 10w + 11$, there's no value of w that produces a true statement when the value of p is 10. It follows that in the given system of equations, the system has no solution when the value of p is 10.

▼ 2.35 — -34

The correct answer is -34. A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the xy-plane are distinct and parallel. Two lines represented by equations in standard form $Ax + By = C$, where A, B, and C are constants, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation. The first equation in the given system can

be written in standard form by subtracting $30y$ from both sides of the equation to yield $48x - 102y = 24$. The second equation in the given system can be written in standard form by adding $16x$ to both sides of the equation to yield $16x + ry = \frac{1}{6}$. The coefficient of x in this second equation, 16, is $\frac{1}{3}$ times the coefficient of x in the first equation, 48. For the lines to be parallel the coefficient of y in the second equation, r, must also be $\frac{1}{3}$ times the coefficient of y in the first equation, -102. Thus, $r = \frac{1}{3}(-102)$, or $r = -34$. Therefore, if the given system has no solution, the value of r is -34.

▼ 2.36 — B

Choice B is correct. Subtracting the second equation, $4x - 8y = 9$, from the first equation, $7x - 5y = 4$, results in $(7x - 5y) - (4x - 8y) = 4 - 9$, or $7x - 5y - 4x + 8y = 5$. Combining like terms on the left-hand side of this equation yields $3x + 3y = -5$.

Choice A is incorrect and may result from miscalculating $4 - 9$ as -13. Choice C is incorrect and may result from miscalculating $4 - 9$ as 5. Choice D is incorrect and may result from adding 9 to 4 instead of subtracting 9 from 4.

▼ 2.37 — D

Choice D is correct. The number of hours Morgan spent running or biking can be calculated by dividing the distance she traveled during that activity by her speed, in miles per hour, for that activity. So the number of hours she ran can be represented by the expression $\frac{r}{5}$, and the number of hours she biked can be represented by the expression $\frac{b}{10}$. It's given that she biked for twice as many hours as she ran, so this can be represented by the equation $\frac{b}{10} = 2\left(\frac{r}{5}\right)$, which can be rewritten as $b = 4r$. It's also given that she ran r miles and biked b miles, and that she ran and biked a total of 200 miles. This can be represented by the equation $r + b = 200$. Substituting $4r$ for b in this equation yields $r + 4r = 200$, or $5r = 200$. Solving for r yields $r = 40$. Determining the number of miles she biked, b, can be found by substituting 40 for r in $r + b = 200$, which yields $40 + b = 200$. Solving for b yields $b = 160$.

Choices A, B, and C are incorrect because they don't satisfy that Morgan biked for twice as many hours as she ran. In choice A, if she biked 80 miles, then she ran 120 miles, which means she biked for 8 hours and ran for 24 hours. In choice B, if she biked 100 miles, then

she ran 100 miles, which means she biked for 10 hours and ran for 20 hours. In choice C, if she biked 120 miles, then she ran for 80 miles, which means she biked for 12 hours and ran for 16 hours.

▼ 2.38 — 20

The correct answer is 20. Adding the first equation to the second equation in the given system yields $5y - 5y = 10x + 5x + 11 - 21$, or $0 = 15x - 10$. Adding 10 to both sides of this equation yields $10 = 15x$. Multiplying both sides of this equation by 2 yields $20 = 30x$. Therefore, the value of $30x$ is 20.

▼ 2.39 — ?

The correct answer is $\frac{1}{2}$. A system of two linear equations has no solution when the graphs of the equations have the same slope and different y-intercepts. Each of the given linear equations is written in the slope-intercept form, $y = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept of the graph of the equation. For these two linear equations, the y-intercepts are $(0,8)$ and $(0,10)$. Thus, if the system of equations has no solution, the slopes of the graphs of the two linear equations must be the same. The slope of the graph of the first linear equation is $\frac{1}{2}$. Therefore, for the system of equations to have no solution, the value of c must be $\frac{1}{2}$. Note that 1/2 and .5 are examples of ways to enter a correct answer.

▼ 2.40 — B

Choice B is correct. A system of two linear equations written in standard form has no solution when the equations are distinct and the ratio of the x-coefficient to the y-coefficient for one equation is equivalent to the ratio of the x-coefficient to the y-coefficient for the other equation. This ratio for the given equation is 2 to 6, or 1 to 3. Only choice B is an equation that isn't equivalent to the given equation and whose ratio of the x-coefficient to the y-coefficient is 1 to 3.

Choice A is incorrect. Multiplying each of the terms in this equation by 2 yields an equation that is equivalent to the given equation. This system would have infinitely many solutions.

Choices C and D are incorrect. The ratio of the x-coefficient to the y-coefficient in $6x-2y=0$ (choice C) is -6 to 2, or -3 to 1. This ratio in $6x+2y=10$ (choice D) is 6 to 2, or 3 to 1. Since neither of these ratios is equivalent to that for the given equation, these systems would have exactly one solution.

▼ 2.41 — D

Choice D is correct. A system of two linear equations in two variables, x and y, has zero solutions if the lines representing the equations in the xy-plane are distinct and parallel. Two lines are distinct and parallel if they have the same slope but different y-intercepts. Each equation in the given system can be written in slope-intercept form $y = mx + b$, where m is the slope of the line representing the equation in the xy-plane and $0, b$ is the y-intercept. Adding $12x$ to both sides of the first equation in the given system of equations, $-12x + 14y = 36$, yields $14y = 12x + 36$. Dividing both sides of this equation by 14 yields $y = \frac{6}{7}x + \frac{18}{7}$. It follows that the first equation in the given system of equations has a slope of $\frac{6}{7}$ and a y-intercept of $0, \frac{18}{7}$. Adding $6x$ to both sides of the second equation in the given system of equations, $-6x + 7y = -18$, yields $7y = 6x - 18$. Dividing both sides of this equation by 7 yields $y = \frac{6}{7}x - \frac{18}{7}$. It follows that the second equation in the given system of equations has a slope of $\frac{6}{7}$ and a y-intercept of $0, -\frac{18}{7}$. Since the slopes of these lines are the same and the y-intercepts are different, it follows that the given system of equations has zero solutions.

Alternate approach: To solve the system by elimination, multiplying the second equation in the given system of equations, $-6x + 7y = -18$, by -2 yields $12x - 14y = 36$. Adding this equation to the first equation in the given system of equations, $-12x + 14y = 36$, yields $(-12x + 12x) + (-14y + 14y) = 36 + 36$, or $0 = 72$. Since this equation isn't true, the given system of equations has zero solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.42 — D

Choice D is correct. Dividing each side of the second equation in the given system by 4 yields $7x + 6y = 5$. It follows that the two equations in the given system are equivalent and any point that lies on the graph of one equation will also lie on the graph of the other

equation. Substituting r for y in the equation $7x + 6y = 5$ yields $7x + 6r = 5$. Subtracting $6r$ from each side of this equation yields $7x = -6r + 5$. Dividing each side of this equation by 7 yields $x = -\frac{6r}{7} + \frac{5}{7}$. Therefore, the point $-\frac{6r}{7} + \frac{5}{7}, r$ lies on the graph of each equation in the xy-plane for each real number r.

Choice A is incorrect. Substituting r for x in the equation $7x + 6y = 5$ yields $7r + 6y = 5$. Subtracting $7r$ from each side of this equation yields $6y = -7r + 5$. Dividing each side of this equation by 6 yields $y = -\frac{7r}{6} + \frac{5}{6}$. Therefore, the point $r, -\frac{7r}{6} + \frac{5}{6}$, not the point $r, -\frac{6r}{7} + \frac{5}{7}$, lies on the graph of each equation.

Choice B is incorrect. Substituting r for x in the equation $7x + 6y = 5$ yields $7r + 6y = 5$. Subtracting $7r$ from each side of this equation yields $6y = -7r + 5$. Dividing each side of this equation by 6 yields $y = -\frac{7r}{6} + \frac{5}{6}$. Therefore, the point $r, -\frac{7r}{6} + \frac{5}{6}$, not the point $r, \frac{7r}{6} + \frac{5}{6}$, lies on the graph of each equation.

Choice C is incorrect. Substituting $\frac{r}{4} + 5$ for x in the equation $7x + 6y = 5$ yields $7\frac{r}{4} + 5 + 6y = 5$, or $\frac{7r}{4} + 35 + 6y = 5$. Subtracting $\frac{7r}{4} + 35$ from each side of this equation yields $6y = -\frac{7r}{4} - 35 + 5$, or $6y = -\frac{7r}{4} - 30$. Dividing each side of this equation by 6 yields $y = -\frac{7r}{24} - 5$. Therefore, the point $\frac{r}{4} + 5, -\frac{7r}{24} - 5$, not the point $\frac{r}{4} + 5, -\frac{r}{4} + 20$, lies on the graph of each equation.

▼ 2.43 — 35

The correct answer is 35. The first equation in the given system of equations defines y as $4x + 1$. Substituting $4x + 1$ for y in the second equation in the given system of equations yields $44x + 1 = 15x - 8$. Applying the distributive property on the left-hand side of this equation yields $16x + 4 = 15x - 8$. Subtracting $15x$ from each side of this equation yields $x + 4 = -8$. Subtracting 4 from each side of this equation yields $x = -12$. Substituting -12 for x in the first equation of the given system of equations yields $y = 4\text{-}12 + 1$, or $y = -47$. Substituting -12 for x and -47 for y into the expression $x - y$ yields $-12 - -47$, or 35.

▼ 2.44 — B

Choice B is correct. Let x represent the total mass, in grams, of the first piece, and let y represent the total mass, in grams, of the second piece. It's given that the sample has a total mass of 50.0 grams. Therefore, the equation $x + y = 50.0$ represents this situation. It's also given that the sample is 50.0% silicon by mass. Therefore, the total mass of the silicon in the sample is $0.50050.0$, or 25.0, grams. It's also given that the first piece was 30.0% silicon by mass and the second piece was 80.0% silicon by mass. Therefore, the masses, in grams, of the silicon in the first and second pieces can be represented by the expressions

$0.300x$ and $0.800y$, respectively. Since the sample was created by combining the first and second pieces, and the total mass of the silicon in the sample is 25.0 grams, the equation $0.300x + 0.800y = 25.0$ represents this situation. Subtracting y from both sides of the equation $x + y = 50.0$ yields $x = 50.0 - y$. Substituting $50.0 - y$ for x in the equation $0.300x + 0.800y = 25.0$ yields $0.30050.0 - y + 0.800y = 25.0$. Distributing 0.300 on the left-hand side of this equation yields $15.0 - 0.300y + 0.800y = 25.0$. Combining like terms on the left-hand side of this equation yields $15.0 + 0.500y = 25.0$. Subtracting 15.0 from both sides of this equation yields $0.500y = 10.0$. Dividing both sides of this equation by 0.500 yields $y = 20.0$. Substituting 20.0 for y in the expression representing the mass, in grams, of the silicon in the second piece, $0.800y$, yields $0.80020.0$, or 16.0. Therefore, the mass, in grams, of the silicon in the second piece is 16.0.

Choice A is incorrect. This is the mass, in grams, of the silicon in the first piece, not the second piece.

Choice C is incorrect. This is the total mass, in grams, of the second piece, not the mass, in grams, of the silicon in the second piece.

Choice D is incorrect. This is the total mass, in grams, of the first piece, not the mass, in grams, of the silicon in the second piece.

▼ 2.45 — D

Choice D is correct. A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the xy-plane are distinct and parallel. The graphs of two lines in the xy-plane represented by equations in the form $Ax + By = C$, where A, B, and C are constants, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation. The first equation in the given system can be written in the form $Ax + By = C$ by subtracting $9y$ from both sides of the equation to yield $4x - 18y = 5$. The second equation in the given system can be written in the form $Ax + By = C$ by subtracting $4x$ from both sides of the equation to yield $-4x + hy = 2$. The coefficient of x in this second equation, -4, is -1 times the coefficient of x in the first equation, 4. For the lines to be parallel, the coefficient of y in the second equation, h, must also be -1 times the coefficient of y in the first equation, -18. Thus, $h = -1(-18)$, or $h = 18$. Therefore, if the given system has no solution, the value of h is 18.

Choice A is incorrect. If the value of h is -9, then the given system would have one solution, rather than no solution.

Choice B is incorrect. If the value of h is 0, then the given system would have one solution, rather than no solution.

Choice C is incorrect. If the value of h is 9, then the given system would have one solution, rather than no solution.

▼ 2.46 — C

Choice C is correct. A system of two linear equations in two variables has at least one solution if the equations represent the same line or lines that intersect at exactly one point. Statement I gives the equation $3x + 13.5y = 10.5$. Multiplying both sides of the given equation, $2x + 9y = 7$, by $\frac{3}{2}$ yields $3x + 13.5y = 10.5$, so the equation in statement I is equivalent to the given equation. Therefore, the system formed by these two equations represents the same line and has infinitely many solutions. Statement II gives the equation $3x - 13.5y = 10.5$. Adding the right- and left-hand sides of $3x + 13.5y = 10.5$ and $3x - 13.5y = 10.5$ yields $3x + 13.5y + 3x - 13.5y = 10.5 + 10.5$, or $6x = 21$. Dividing both sides of this equation by 6 yields $x = \frac{21}{6}$. Therefore, the lines represented by these equations intersect at exactly one point, where $x = \frac{21}{6}$, and the system formed by the given equation and the equation in statement II has one solution. It follows that both the equation in statement I and the equation in statement II could be the other equation in a system of equations with at least one solution.

Choice A is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.

Choice B is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.

Choice D is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.

▼ 2.47 — D

Choice D is correct. A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the xy-plane are parallel and distinct. Lines represented by equations in standard form, $Ax + By = C$ and $Dx + Ey = F$, are parallel and distinct if the coefficients of x and y in one equation are proportional to the corresponding coefficients in the other equation, meaning $\frac{D}{A} = \frac{E}{B}$, and the constants are not proportional, meaning $\frac{F}{C}$ is not equal to $\frac{D}{A}$ nor $\frac{E}{B}$. It's given that one of the equations in a system of two linear equations is $38(x - n) = 38y + 38n$, where n is a positive constant.

Distributing 38 on the left-hand side of this equation yields $38x - 38n = 38y + 38n$. Adding $38n$ to both sides of this equation yields $38x = 38y + 76n$. Subtracting $38y$ from both sides of this equation yields $38x - 38y = 76n$. Therefore, the given equation can be written in the form $Ax + By = C$, where $A = 38$, $B = -38$, and $C = 76n$. The equation in choice D, $4x - 4y = 4n$, is written in the form $Dx + Ey = F$, where $D = 4$, $E = -4$, and $F = 4n$. Using these values, it follows that the value of $\frac{D}{A}$ is $\frac{4}{38}$, or $\frac{2}{19}$; the value of $\frac{E}{B}$ is $\frac{-4}{-38}$, or $\frac{2}{19}$; and the value of $\frac{F}{C}$ is $\frac{4n}{76n}$, or $\frac{1}{19}$, since n is a positive constant. Since $\frac{D}{A} = \frac{E}{B}$ and since $\frac{F}{C}$ is not equal to $\frac{D}{A}$ nor $\frac{E}{B}$, it follows that the lines represented by these two equations are parallel and distinct. Therefore, the second equation in this system of equations with no solution could be $4x - 4y = 4n$.

Choice A is incorrect. The system consisting of this equation and the given equation has infinitely many solutions rather than no solution.

Choice B is incorrect. The system consisting of this equation and the given equation has one solution rather than no solution.

Choice C is incorrect. The system consisting of this equation and the given equation has one solution rather than no solution.

▼ 2.48 — B

Choice B is correct. The two given equations are equivalent because the second equation can be obtained from the first equation by multiplying each side of the equation by 5. Thus, the graphs of the equations are coincident, so if a point lies on the graph of one of the equations, it also lies on the graph of the other equation. A point x, y lies on the graph of an equation in the xy-plane if and only if this point represents a solution to the equation. It is sufficient, therefore, to find the point that represents a solution to the first given equation. Substituting the x- and y-coordinates of choice B, $-\frac{3r}{2} + \frac{7}{2}$ and r, for x and y, respectively, in the first equation yields $2\left(-\frac{3r}{2} + \frac{7}{2}\right) + 3r = 7$, which is equivalent to $-3r + 7 + 3r = 7$, or $7 = 7$. Therefore, the point $\left(-\frac{3r}{2} + \frac{7}{2}, r\right)$ represents a solution to the first equation and thus lies on the graph of each equation in the xy-plane for the given system.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.49 — .2857, 2/7

The correct answer is $\frac{2}{7}$. It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions if and only if the two linear equations are equivalent. Multiplying each side of the first equation in the system by $\frac{35}{4}$ yields $\frac{35}{4}\frac{2}{5}x + \frac{7}{5}y = \frac{35}{4}\frac{2}{7}$, or $\frac{7}{2}x + \frac{49}{4}y = \frac{5}{2}$. Since this equation is equivalent to the second equation and has the same right side as the second equation, the coefficients of x and y, respectively, should also be the same. It follows that $g = \frac{7}{2}$ and $k = \frac{49}{4}$. Therefore, the value of $\frac{g}{k}$ is $\frac{\frac{7}{2}}{\frac{49}{4}}$, or $\frac{2}{7}$. Note that 2/7, .2857, 0.285, and 0.286 are examples of ways to enter a correct answer.

▼ 2.50 — D

Choice D is correct. A system of two linear equations in two variables, x and y, has exactly one solution if the graphs of the lines represented by the equations in the xy-plane have different slopes. In the xy-plane, the slope of a line defined by an equation in the form $Ax + By = C$, where A, B, and C are constants, is $-\frac{A}{B}$. It's given that one equation in the system is $x + y = 20$. It follows that for this equation, $A = 1$ and $B = 1$. Therefore, the value of $-\frac{A}{B}$ is $-\frac{1}{1}$, or -1. For the equation in choice D, $x - y = 20$, $A = 1$ and $B = -1$. Therefore, the value of $-\frac{A}{B}$ is $-\frac{1}{-1}$, or 1. Since the slopes are different, it follows that the system of $x + y = 20$ and $x - y = 20$ has exactly one solution. Thus, the second equation in the system could be $x - y = 20$.

Choice A is incorrect. The equation $3x + 3y = 60$ can be rewritten as $x + y = 20$. Since $x + y = 20$ is the same as the given equation, the system of $3x + 3y = 60$ and $x + y = 20$ has infinitely many solutions rather than exactly one solution.

Choice B is incorrect. The equation $-x - y = 40$ can be rewritten as $x + y = -40$. The graph of this line has a slope of -1. Since the graphs of the lines represented by the equations $x + y = -40$ and $x + y = 20$ have the same slope in the xy-plane and the equations aren't equivalent, this system has no solution rather than exactly one solution.

Choice C is incorrect. The equation $3x + 3y = 14$ can be rewritten as $x + y = \frac{14}{3}$. The graph of this line has a slope of -1. Since the graphs of the lines represented by the equations $x + y = \frac{14}{3}$ and $x + y = 20$ have the same slope in the xy-plane and the equations aren't equivalent, this system has no solution rather than exactly one solution.

▼ 2.51 — 3.5, 7/2

The correct answer is $\frac{7}{2}$. A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the xy-plane are distinct and parallel. Two lines represented by equations in standard form $Ax + By = C$, where A, B, and C are

constants, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation. The first equation in the given system, $\frac{7}{8}y - \frac{5}{8}x = \frac{4}{7} - \frac{7}{8}y$, can be written in standard form by adding $\frac{7}{8}y$ to both sides of the equation, which yields $\frac{14}{8}y - \frac{5}{8}x = \frac{4}{7}$, or $-\frac{5}{8}x + \frac{14}{8}y = \frac{4}{7}$. Multiplying each term in this equation by -8 yields $5x - 14y = -\frac{32}{7}$. The second equation in the given system, $\frac{5}{4}x + \frac{7}{4} = py + \frac{15}{4}$, can be written in standard form by subtracting $\frac{7}{4}$ and py from both sides of the equation, which yields $\frac{5}{4}x - py = \frac{8}{4}$. Multiplying each term in this equation by 4 yields $5x - 4py = 8$. The coefficient of x in the first equation, $5x - 14y = -\frac{32}{7}$, is equal to the coefficient of x in the second equation, $5x - 4py = 8$. For the lines to be parallel, and for the coefficients for x and y in one equation to be proportional to the corresponding coefficients in the other equation, the coefficient of y in the second equation must also be equal to the coefficient of y in the first equation. Therefore, $-14 = -4p$. Dividing both sides of this equation by -4 yields $\frac{-14}{-4} = p$, or $p = \frac{7}{2}$. Therefore, if the given system of equations has no solution, the value of p is $\frac{7}{2}$. Note that 7/2 and 3.5 are examples of ways to enter a correct answer.

▼ 2.52 — 25

The correct answer is 25. It's given that a piece of wire has a length of 32 inches and is cut into two parts. It's also given that one part has a length of x inches and the other part has a length of y inches. It follows that the equation $x + y = 32$ represents this situation. It's also given that the value of x is 4 more than 3 times the value of y, or $x = 3y + 4$. Substituting $3y + 4$ for x in the equation $x + y = 32$ yields $3y + 4 + y = 32$. Combining like terms on the left-hand side of this equation yields $4y + 4 = 32$. Subtracting 4 from both sides of this equation yields $4y = 28$. Dividing both sides of this equation by 4 yields $y = 7$. Substituting 7 for y in the equation $x = 3y + 4$ yields $x = 3(7) + 4$, or $x = 25$. Therefore, the value of x is 25.

▼ 2.53 — -28

The correct answer is -28. A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the xy-plane are distinct and parallel. The graphs of two lines in the xy-plane represented by equations in the form $Ax + By = C$, where A, B, and C are constants, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients for x and y in the other equation. The first equation in the given system, $48x - 64y = 48y + 24$, can be written in the form $Ax + By = C$ by subtracting $48y$ from both sides of the equation to yield $48x - 112y = 24$. The second equation in the given system, $ry = \frac{1}{8} - 12x$, can be written in the form $Ax + By = C$ by adding $12x$ to both sides of the equation to yield $12x + ry = \frac{1}{8}$. The coefficient of x in the second equation is $\frac{1}{4}$ times the coefficient of x in the first equation. That is, $48\frac{1}{4} = 12$. For the lines to be

parallel, the coefficient of y in the second equation must also be $\frac{1}{4}$ times the coefficient of y in the first equation. Therefore, $-112\frac{1}{4} = r$, or $-28 = r$. Thus, if the given system has no solution, the value of r is -28.

▼ 2.54 — ?

The correct answer is $\frac{3}{2}$. One method for solving the system of equations for y is to add corresponding sides of the two equations. Adding the left-hand sides gives $(-x+y)+(x+3y)$, or 4y. Adding the right-hand sides yields $-3.5+9.5=6$. It follows that $4y=6$. Finally, dividing both sides of $4y=6$ by 4 yields $y=\frac{6}{4}$ or $\frac{3}{2}$. Note that 3/2 and 1.5 are examples of ways to enter a correct answer.

▼ 2.55 — ?

The correct answer is 180. Let a be the number of adult tickets sold and c be the number of child tickets sold. Since the theater sold a total of 30 tickets, a + c = 30. The price per adult ticket is $12, and the price per child ticket is $8. Since the theater received a total of $300 for the 30 tickets sold, it follows that 12a + 8c = 300. To eliminate c, the first equation can be multiplied by 8 and then subtracted from the second equation:

$$\begin{array}{r} 12a+8c = 300 \\ -8a-8c = -240 \\ \hline 4a+0c=60 \end{array}$$

Because the question asks for the amount spent on adult tickets, which is 12a dollars, the resulting equation can be multiplied by 3 to give 3(4a) = 3(60) = 180. Therefore, $180 was spent on adult tickets.

Alternate approach: If all the 30 tickets sold were child tickets, their total price would be 30($8) = $240. Since the actual total price of the 30 tickets was $300, the extra $60 indicates that a certain number of adult tickets, a, were sold. Since the price of each adult ticket is $4 more than each child ticket, 4a = 60, and it follows that 12a = 180.

▼ 2.56 — D

Choice D is correct. A system of two linear equations has no solution when the graphs of the equations have the same slope and different y-coordinates of the y-intercepts. Each of the given equations is written in the slope-intercept form of a linear equation, $y = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept of the graph of the equation. For these two linear equations, the y-coordinates of the y-intercepts are different: 1 and -8. Thus, if the system of equations has no solution, the slopes of the two linear equations must be the same. The slope of the first linear equation is 2. Therefore, for the system of equations to have no solution, the value of a must be 2.

Choices A, B, and C are incorrect and may result from conceptual and computational errors.

▼ 2.57 — D

Choice D is correct. A system of two linear equations has infinitely many solutions if one equation is equivalent to the other. This means that when the two equations are written in the same form, each coefficient or constant in one equation is equal to the corresponding coefficient or constant in the other equation multiplied by the same number. The equations in the given system of equations are written in the same form, with x and y on the left-hand side and a constant on the right-hand side of the equation. The coefficient of y in the second equation is equal to the coefficient of y in the first equation multiplied by 3. Therefore, a, the coefficient of x in the second equation, must be equal to 3 times the coefficient of x in the first equation: $a = (\frac{1}{2})(3)$, or $a = \frac{3}{2}$.

Choices A, B, and C are incorrect. When $a = -\frac{1}{2}$, $a = 0$, or $a = \frac{1}{2}$, the given system of equations has one solution.

▼ 2.58 — 3

The correct answer is 3. Adding the second equation to the first equation in the given system of equations yields $28x - 28x + 47y + 47y = 12 + 12$, or $87y = 24$. Dividing both sides of this equation by 8 yields $7y = 3$. Substituting 3 for $7y$ in the first equation, $28x + 47y = 12$, yields $28x + 43 = 12$, or $28x + 12 = 12$. Subtracting 12 from both sides of this equation yields $28x = 0$. Dividing both sides of this equation by 2 yields $8x = 0$. Substituting 0 for $8x$ and 3 for $7y$ in the expression $8x + 7y$ yields $0 + 3$, or 3. Therefore, the value of $8x + 7y$ is 3.

▼ 2.59 — 8

The correct answer is 8. The given system of equations can be solved using the elimination method. Multiplying both sides of the second equation in the given system by -2 yields $-2ty = -1 - 4x$, or $-1 - 4x = -2ty$. Adding this equation to the first equation in the given system, $4x - 6y = 10y + 2$, yields $4x - 6y + -1 - 4x = 10y + 2 + -2ty$, or $-1 - 6y = 10y - 2ty + 2$. Subtracting $10y$ from both sides of this equation yields $-1 - 6y - 10y = 10y - 2ty + 2 - 10y$, or $-1 - 16y = -2ty + 2$. If the given system has no solution, then the equation $-1 - 16y = -2ty + 2$ has no solution. If this equation has no solution, the coefficients of y on each side of the equation, -16 and $-2t$, must be equal, which yields the equation $-16 = -2t$. Dividing both sides of this equation by -2 yields $8 = t$. Thus, if the system has no solution, the value of t is 8.

Alternate approach: A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the xy-plane are parallel and distinct. Lines represented by equations in the form $Ax + By = C$, where A, B, and C are constant terms, are parallel if the ratio of the x-coefficients is equal to the ratio of the y-coefficients, and distinct if the ratio of the x-coefficients are not equal to the ratio of the constant terms. Subtracting $10y$ from both sides of the first equation in the given system yields $4x - 6y - 10y = 10y + 2 - 10y$, or $4x - 16y = 2$. Subtracting $2x$ from both sides of the second equation in the given system yields $ty - 2x = \frac{1}{2} + 2x - 2x$, or $-2x + ty = \frac{1}{2}$. The ratio of the x-coefficients for these equations is $-\frac{2}{4}$, or $-\frac{1}{2}$. The ratio of the y-coefficients for these equations is $-\frac{t}{16}$. The ratio of the constant terms for these equations is $\frac{\frac{1}{2}}{2}$, or $\frac{1}{4}$. Since the ratio of the x-coefficients, $-\frac{1}{2}$, is not equal to the ratio of the constants, $\frac{1}{4}$, the lines represented by the equations are distinct. Setting the ratio of the x-coefficients equal to the ratio of the y-coefficients yields $-\frac{1}{2} = -\frac{t}{16}$. Multiplying both sides of this equation by -16 yields $-\frac{1}{2}-16 = -\frac{t}{16}-16$, or $t = 8$. Therefore, when $t = 8$, the lines represented by these equations are parallel. Thus, if the system has no solution, the value of t is 8.

▼ 2.60 — 1.8, 9/5

The correct answer is $\frac{9}{5}$. Multiplying the first equation in the given system by 2 yields $10x + 28y = 90$. Subtracting the second equation in the given system, $10x + 7y = 27$, from $10x + 28y = 90$ yields $10x + 28y - 10x + 7y = 90 - 27$, which is equivalent to $10x + 28y - 10x - 7y = 63$, or $21y = 63$. Dividing both sides of this equation by 21 yields $y = 3$. The value of x can be found by substituting 3 for y in either of the two given equations. Substituting 3 for y in the equation $10x + 7y = 27$ yields $10x + 73 = 27$, or $10x + 21 = 27$. Subtracting 21 from both sides of this equation yields $10x = 6$. Dividing both sides of this

equation by 10 yields $x = \frac{6}{10}$, or $x = \frac{3}{5}$. Therefore, the value of xy is $\frac{3}{5}3$, or $\frac{9}{5}$. Note that 9/5 and 1.8 are examples of ways to enter a correct answer.

▼ 2.61 — A

Choice A is correct. Dividing both sides of the second equation in the given system by 3 yields $8x + 7y = 9$, which is the first equation in the given system. Therefore, the first and second equations represent the same line in the xy-plane. If the x- and y-coordinates of a point satisfy an equation, the point lies on the graph of the equation in the xy-plane. Choice A is a point with x-coordinate r and y-coordinate $-\frac{8r}{7} + \frac{9}{7}$. Substituting r for x and $-\frac{8r}{7} + \frac{9}{7}$ for y in the equation $8x + 7y = 9$ yields $8r + 7-\frac{8}{7}r + \frac{9}{7} = 9$. Applying the distributive property to the left-hand side of this equation yields $8r - 8r + 9 = 9$. Combining like terms on the left-hand side of this equation yields $9 = 9$, so the coordinates of the point $r, -\frac{8}{7}r + \frac{9}{7}$ satisfy both equations in the given system. Therefore, for each real number r, the point $r, -\frac{8}{7}r + \frac{9}{7}$ lies on the graph of each equation in the xy-plane for the given system.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.62 — 11

The correct answer is 11. It's given that the graphs of the equations in the given system intersect at the point $(q, 19)$, where q is a constant. Therefore, the coordinates of this point must satisfy both equations. Substituting the point $(q, 19)$ into the first equation, $-x - wy = -337$, yields $-q - w(19) = -337$. Adding $19w$ to both sides of this equation yields $-q = -337 + 19w$, which is equivalent to $q = 337 - 19w$. Substituting the point $(q, 19)$ into the second equation yields $2q - w(19) = 47$. Substituting $337 - 19w$ in place of q in the equation $2q - w(19) = 47$ yields $2(337 - 19w) - 19w = 47$. Applying the distributive property to the left-hand side of this equation yields $674 - 38w - 19w = 47$. Combining like terms on the left-hand side of this equation yields $674 - 57w = 47$. Subtracting 674 from both sides of this equation yields $-57w = -627$. Dividing both sides of this equation by -57 yields $w = 11$.

▼ 2.63 — 6

The correct answer is 6. A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the xy-plane are parallel and distinct. Lines represented by equations in standard form, $Ax + By = C$ and $Dx + Ey = F$, are parallel if

the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation, meaning $\frac{D}{A} = \frac{E}{B}$; and the lines are distinct if the constants are not proportional, meaning $\frac{F}{C}$ is not equal to $\frac{D}{A}$ or $\frac{E}{B}$. The first equation in the given system is $\frac{3}{2}y - \frac{1}{4}x = \frac{2}{3} - \frac{3}{2}y$. Multiplying each side of this equation by 12 yields $18y - 3x = 8 - 18y$. Adding $18y$ to each side of this equation yields $36y - 3x = 8$, or $-3x + 36y = 8$. The second equation in the given system is $\frac{1}{2}x + \frac{3}{2} = py + \frac{9}{2}$. Multiplying each side of this equation by 2 yields $x + 3 = 2py + 9$. Subtracting $2py$ from each side of this equation yields $x + 3 - 2py = 9$. Subtracting 3 from each side of this equation yields $x - 2py = 6$. Therefore, the two equations in the given system, written in standard form, are $-3x + 36y = 8$ and $x - 2py = 6$. As previously stated, if this system has no solution, the lines represented by the equations in the xy-plane are parallel and distinct, meaning the proportion $\frac{1}{-3} = \frac{-2p}{36}$, or $-\frac{1}{3} = -\frac{p}{18}$, is true and the proportion $\frac{6}{8} = \frac{1}{-3}$ is not true. The proportion $\frac{6}{8} = \frac{1}{-3}$ is not true. Multiplying each side of the true proportion, $-\frac{1}{3} = -\frac{p}{18}$, by -18 yields $6 = p$. Therefore, if the system has no solution, then the value of p is 6.